

MediCare – Online Pharmacy and Medicine Ordering System

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❖ ABSTRACT:

MediCare is an online pharmacy platform that enables customers to browse medicines, upload prescriptions, place orders and track deliveries through a responsive React front-end. Secure JWT authentication protects user sessions while Node.js and Express handle API calls, business logic and communication with MongoDB. Pharmacists access an admin dashboard to manage inventory, verify prescriptions and process orders. File uploads are managed by Multer and notifications are sent to pharmacists when new orders arrive. The system aims to streamline the medicine ordering process, reduce errors and improve accessibility for users across the region.

❖ INTRODUCTION OF THE PROJECT:

The growing demand for digital healthcare services has highlighted the need for a reliable online pharmacy system. MediCare addresses this by combining a modern React UI with a robust Node.js/Express back-end and MongoDB storage. Users can safely search for medicines, upload doctor prescriptions and receive real-time order updates. Pharmacists benefit from a dedicated dashboard to control stock levels, validate prescriptions and dispatch orders efficiently. By leveraging JWT for authentication and Multer for secure file handling, the application ensures data privacy and smooth interaction between customers and pharmacy staff.

❖ **PROJECT OBJECTIVES:**

- Enable customers to search and filter medicines quickly using a responsive UI.
- Allow secure upload and storage of prescription images for pharmacist review.
- Provide a JWT-based authentication system for protected user sessions.
- Give pharmacists an admin panel to manage inventory and process orders.
- Implement real-time order tracking and notification for both customers and staff.

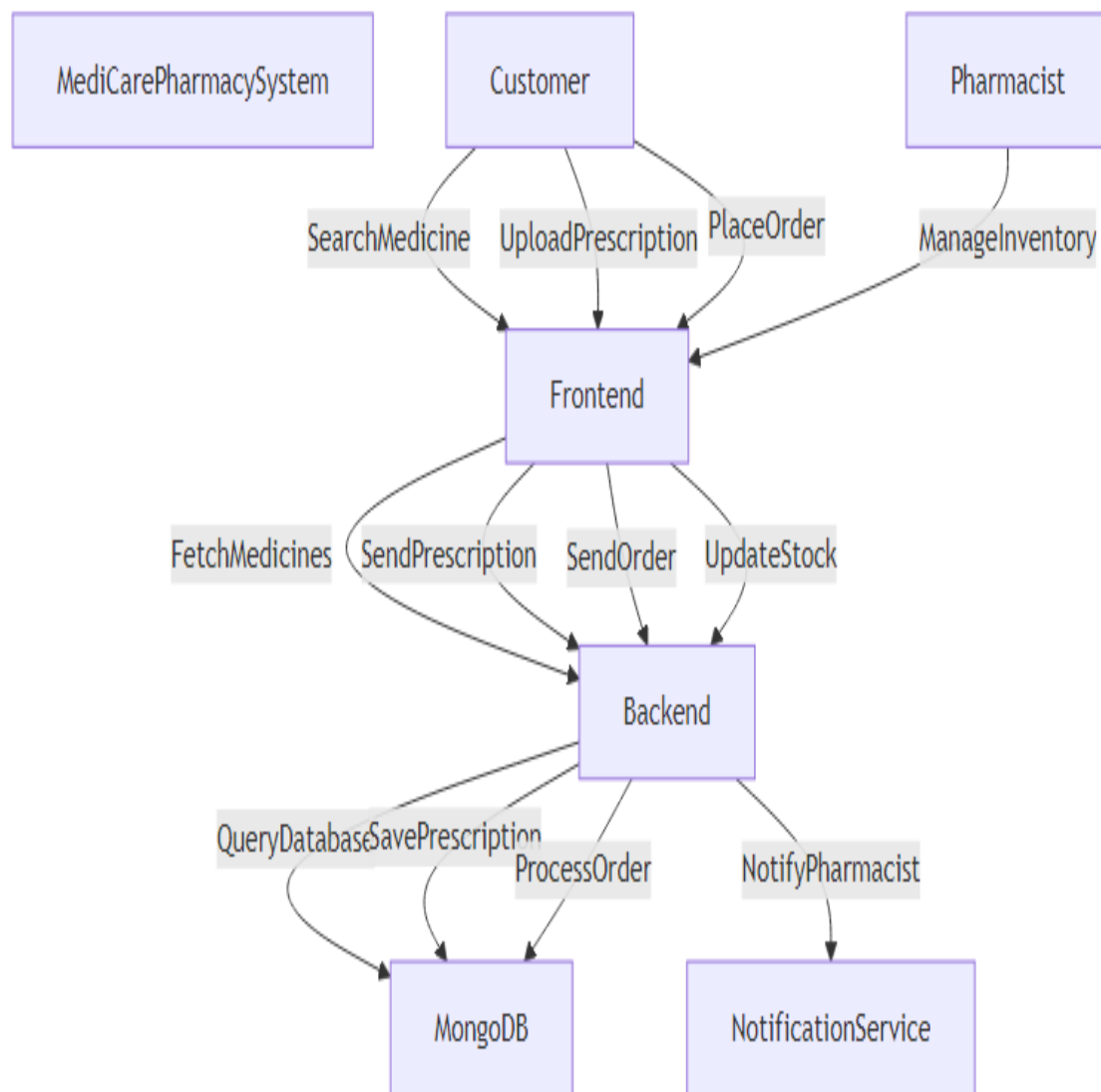
❖ **PROBLEM OF THE STATEMENT:**

Many patients still rely on physical visits to pharmacies, which can be time-consuming and risky, especially during pandemics. Existing online pharmacies often lack integrated prescription handling, secure authentication, and real-time inventory updates, leading to errors, delays and poor user experience. MediCare aims to close these gaps by offering a unified platform where customers can search medicines, upload verified prescriptions, place orders and track deliveries, while pharmacists manage stock and process orders seamlessly through a secure admin dashboard.

❖ **MOTIVATION OF THE PROJECT:**

- Reduce the need for in-person pharmacy visits and associated waiting times.
- Improve accuracy of prescription validation through digital uploads.
- Provide 24/7 access to medicine ordering for users in remote areas.
- Streamline inventory management for pharmacists with real-time data.
- Enhance security and privacy of user data using JWT tokens.

❖ **BLOCK DAIGRAM OF PROJECT:**



❖ **WORKING OF THE PROJECT:**

A customer uses the React front-end to search medicines, which triggers an API call to the Express back-end. The back-end queries MongoDB and returns matching items. When a prescription is uploaded, Multer stores the file and its reference is saved in the database. The customer places an order, the back-end creates an order record, updates stock and notifies the pharmacist via the notification service. Pharmacists log into the admin dashboard, verify prescriptions, adjust inventory and mark orders as dispatched, while the customer can track delivery status in real time.

❖ **SOFTWARE REQUIREMENT:**

1. Front-end:

1. React 18 with functional components and hooks
2. Axios for API communication and JWT handling

2. Back-end:

1. Node.js with Express framework
2. Multer for file uploads, jsonwebtoken for authentication

3. Database Services:

1. MongoDB Atlas for cloud-hosted document storage
2. Notification service (e.g., Socket.io or email SMTP)

❖ **ADVANTAGES OF THE PROJECT:**

- The system offers a single place for customers to search, order and track medicines, eliminating the need to visit multiple pharmacies and saving valuable time.
- Secure JWT authentication ensures that user sessions are protected against tampering, providing confidence in data privacy for both patients and pharmacists.
- Prescription images are stored safely using Multer and MongoDB, allowing pharmacists to verify medication needs without requiring physical documents.
- Real-time inventory updates prevent stock-outs and enable pharmacists to manage supplies efficiently, reducing waste and improving revenue.
- The admin dashboard gives pharmacists a clear overview of orders, prescriptions and stock levels, streamlining workflow and reducing manual errors.
- Scalable architecture based on React and Node.js allows easy addition of new features such as mobile apps or AI-driven drug suggestions in the future.

❖ **APPLICATIONS OF THE PROJECT:**

- Online medicine ordering for urban and rural populations.
- Remote prescription verification by licensed pharmacists.
- Integration with delivery services for home medication delivery.
- Patient portals for tracking order history and refill reminders.
- Pharmacy inventory management across multiple branches.
- Health-care platforms that need a secure pharmacy module.

❖ **REFERENCES:**

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- [6] Node.js Official Site <https://nodejs.org>
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